Unlocking 5G Network Slicing: A Comprehensive Survey on Blockchain Marketplace utilizing NFTs, AI, and Advanced Resource Management

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***Abstract*—This survey paper explores the incorporation of blockchain with 5G networks for network slicing, resource allocation, and security. It is argued how blockchain, in combination with AI and NFTs, can automate and optimize 5G network operations with transparency and security. The paper presents a comparative analysis of different blockchain-based 5G solutions highlighting the advantages of resource efficiency, scalability, and security. It also covers the critical challenges faced with the concept, which include increased computational demands, integration complexity, and network latency, and emphasizes the continuation of research to overcome them.**

***Index Terms*—Blockchain, 5G, Marketplace, Network-Slicing**

I. Introduction

Network slicing represents a pivotal advancement in telecommunications, particularly with the advent of 5G technology. It enables the division and allocation of resources efficiently across various virtual networks that share a common physical infrastructure [1]. This allows for highly specialized virtual networks that cater to specific demands such as speed or capacity, offering flexibility and cost savings through network softwarization. By leveraging Software-Defined Networking (SDN) and Network Function Virtualization (NFV), network slicing supports the creation of customizable, end-to-end (E2E) slices, ensuring consistent service quality across multiple administrative domains [2].

In combination with these advancements, blockchain technology plays a crucial role in automating complex corporate processes within 5G networks. Blockchain enhances resource allocation and overall network efficiency through its decentralized and secure framework [3]. Additionally, the integration of Non-Fungible Tokens (NFTs) into blockchain-based 5G networks enables unique digital representations of network slices, ensuring improved transparency and security in resource management. However, challenges such as increased computational demands and potential latency issues remain, posing obstacles to the seamless incorporation of blockchain in 5G networks [1]. Blockchain technology enhances corporate processes in 5G networks, particularly by boosting resource allocation and network efficiency [4] [5]. The use of Non-Fungible Tokens (NFTs) allows network slices to be represented as unique digital assets, improving transparency and security in resource management. Studies show that blockchain integration in 5G supports high-speed data transfer and secure transactions, which is particularly beneficial for industries with stringent security requirements [6]. However, challenges like computational demands and network latency remain, potentially impacting service quality [4].

In Industry 4.0, the IIoT requires rapid data collection and processing, linking billions of devices. Yet, centralized

architectures pose challenges such as access control and single points of failure [7]. Blockchain offers a solution through its decentralized and transparent ledger, which, when combined with 5G, can securely handle vast amounts of data [8].

As 5G and beyond (B5G) networks evolve, blockchain's role in optimizing network slicing continues to grow. Research on combining AI and edge computing with blockchain indicates significant efficiency improvements [9], [10]. Integrating AI into Open RAN architectures within B5G systems also enhances flexibility and modularity, enabling more autonomous network operations [11].

In summary, the integration of blockchain technology with 5G networks revolutionizes contemporary telecommunications, significantly enhancing security, efficiency, and resource management. Ongoing research highlights the potential of combining artificial intelligence with blockchain and advanced network architectures to drive progress in industrial automation, intelligent manufacturing, and various other domains. The following sections outline the structure of this survey: Section II presents the fundamental concepts of 5G and blockchain technology, Section III explores existing solutions and their comparative analysis, Section IV discusses the main findings and remaining challenges, and finally, Section V concludes this survey paper.

II. Background

Before delving into the examined proposals, we need to clarify some fundamental terms related to 5G networking and blockchain technology. This section provides concise definitions and explanations to establish a clear understanding of these concepts.

a) Radio Access Network: The Radio Access Network (RAN) comprises the logical components between the receiver and the core network in end-to-end telecommunications, with communication facilitated via interfaces. Open RAN architecture aims to foster a multi-vendor ecosystem, introducing flexibility and promoting innovation. It includes dedicated components for intelligent applications that control RAN communications, supporting new use cases such as the tactile internet and autonomous driving. As we transition to 6G networks, the increasingly complex topology will present challenges in service management and deployment [12]. Network softwarization, realized through SDN, NFV, and Network Slicing (NS), is crucial in this context. A network slice is a virtual network that allocates services over a shared infrastructure, providing tailored resources and isolation [11].

b) Blockchain: In this context, blockchain technology enhances network slicing and resource allocation by enabling marketplaces for the buying, selling, and trading of goods, services, or digital assets. These platforms leverage blockchain's transparency and security, allowing peer-to-peer (P2P) transactions that minimize costs [13]. Smart contracts, which are self-executing and have terms embedded in code, manage marketplace logic, ensuring transparency, security, and efficient execution across various applications, including finance, supply chain management, and legal agreements [14].

Marketplace smart contracts will oversee the management of network resources represented as Non-Fungible Tokens (NFTs), which enhance the authenticity and provenance of digital assets such as art [15], music [16], and virtual real estate [17]. By elevating network slices to digital asset status, this approach leverages NFTs' uniqueness, immutability, and transparent tracking. To facilitate the creation and management of tokens, decentralized applications (DApps) run on blockchain networks, offering resilience, transparency, and security while reducing censorship and downtime risks [18]. Their open-source nature makes them suitable for various sectors, including finance (DeFi) [19], gaming [20], social media [21], supply chain management [22], and education [23].

c) Blockchain Applications in Network Slicing: Integrating blockchain technology into network slicing automation, enhances efficiency, security and enables innovative business models, leading to improved service delivery and user satisfaction. Additionally, private blockchain networks provide improved security by ensuring that sensitive information is accessible only to authorized entities. It also facilitates new business models based on Distributed Ledger Technology (DLT), improving profitability for providers and cost-effectiveness for consumers.

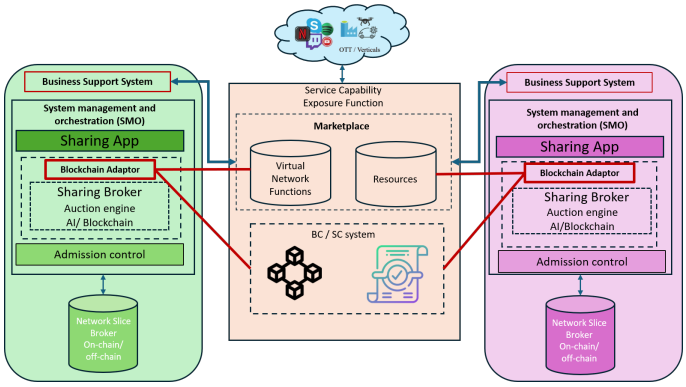

Fig. 1: Abstract Concept and Methodology for a Blockchain-Integrated Marketplace in 5G Network Slices

The Figure 1 presents a generic, abstract overview of a blockchain-enabled framework for managing network slices in 5G systems. It provides a high-level depiction of how the main concepts discussed in this survey are interconnected. At the core of the abstract concept is a marketplace, which manages access to Virtual Network Functions (VNFs) and other essential resources. The Sharing App, positioned within the System Management and Orchestration (SMO) layer, plays a key role in orchestrating network slicing with the help of artificial intelligence (AI), blockchain, and an auction engine.

The diagram highlights key components, such as the Network Slice Broker, which operates both on-chain and off-chain

to facilitate resource allocation via the marketplace. Secure and transparent transactions are managed through the Blockchain Adaptor, linking internal sharing mechanisms in the SMO with the blockchain and smart contract system (BC/SC). The arrows indicate the flow of data and the interaction between these components, showing how the Sharing App connects with the blockchain to automate decision-making processes.

Overall, this interconnected abstract concept provides a decentralized and efficient approach to network resource management, which is aligned with the concepts of AI-driven optimization and blockchain-secured transactions. The ideas outlined in this survey are closely reflected in the abstract structure of this figure, which acts as a simplified representation of the complex systems being discussed.

III. Comparative Analysis of Existing Solutions

Previous research studies have focused on incorporating NFTs to represent network slices. In this direction, the papers show the integration between AI and blockchain that creates possibilities for optimizing network performance. However, challenges such as scalability, integration complexity, and cybersecurity are significant obstacles that remain for their full potential to be tapped. This section will review the main contributions and ongoing research efforts. The discussed concepts are categorized in two main sections. The first includes the papers that integrade NFTs in their implementation and the second the papers that focus on different technologies than NFTs.

A. With the use of NFTs

Combining blockchain and 5G networks with NFTs embeds a new approach to network slice management and representation. The following section elaborates on how NFTs aim to revolutionize network slicing and resource management in the 5G networks toward advanced industrial and heightened telecommunication applications.

Bandara et al. proposed a framework named Kaputa [24]. It is a blockchain framework that utilizes NFTs and model cards for 5G and 6G, whereby network slices are represented and traded within a Moose blockchain [25]. Additionally, Kaputa implements a dynamic NFT marketplace that allows for flexible and rapid provisioning of network slices. Customers can purchase network slices based on specific requirements. This model not only provides a new revenue stream for network providers but also enhances customer satisfaction by offering tailored network services on demand. However, challenges include managing NFT transaction complexity and ensuring security, especially with large data volumes and privacy concerns. Kaputa manages these challenges through off-chain storage of extensive data and using blockchain for transactional integrity.

Foytik et al. proposed SliceGPT [26], which integrates GPT-3.5 large language models (LLM) with blockchain and NFTs to actualize an intelligent network slice broker and marketplace. Powered by the Ethereum blockchain for managing smart contracts and NFTs, SliceGPT is AI-enabled to impute and optimize the network slices based on data-driven insights. The transactions use smart contracts, which helps manage the network slices. SliceGPT utilizes the on-chain and off-chain combination for keeping important transaction data, smart contracts, and NFT. Nevertheless, it stores big data, like network slice templates and other massive datasets off-chain, to effectively lessen the load on the blockchain. A slice would be realized by an NFT that represented it. However, managing network slicing relies on on-chain mechanisms, utilizing identity contracts for verification, slice contracts for resource allocation, and NFT contracts for the creation and management of network slices. Additionally, a model card contract stores detailed information about each slice. Despite these advancements, challenges remain, particularly concerning the high computational demands and privacy concerns associated with AI data processing.

Shetty et al. proposed Federated Learning concepts which utilizes Model Cards for 5G/6G Network Sliced Environment [27]. It integrates technologies that guarantee functionality within a network-sliced 5G/6G infrastructure. The implementation is based on NFTs that are traded on the Cosmos blockchain within a federated learning framework. Cosmos is considered the ”Internet of Blockchains” due to its architecture, while it also utilizes a Tendermint consensus algorithm [28] created for all-round fast and secure transactions. The system ensures that data related to Unmanned Aerial Vehicle (UAV) surveillance remains private; this is achieved by processing the data locally at the UAVs themselves. Data are locally processed in each UAV and not sent further away over the network, preventing any breach. In this way, federated learning will not share raw data; instead, it allows a UAV to train models locally based on their corresponding data and then share only model updates. Moreover, the system is able to monitor and confirm the process of ownership and the history of transactions in a transparent manner, hence adding an extra layer of trust. Transactions and setups are configured so that information is accessed by authorized entities only and in such a way that the security features of NFTs are further leveraged. However, the implementation faces challenges concerning to scalability and security.

Michoulis et al. introduces FraMark [29], a blockchain-based framework for 5G network management utilizing fractional NFTs (F-NFTs) and Hyperledger Fabric. This feature tackles the large challenges in 5G resource allocation, enabling the shared ownership of dynamic leasing of network slices with F-NFTs, which is impossible by traditional NFTs. FraMark introduces a marketplace where subdivided portions of network slices will be represented by F-NFTs; thus, it allows secure and facilitated trading. With Hyperledger Fabric, it runs low latency, high transaction throughput, and resource allocation securely within the network. Evidence can be obtained by analyzing a number of experiments on the utilization of CPUs and memory, latency analysis, and block size assessments that prove the

effectiveness of the system in optimization and thereby improvement of general network efficiency. The implementation of a platform architecture, combined with smart contracts and a secure marketplace, provides the ability to efficiently manage resources dynamically and positions it as one of the most promising and flexible solutions in the 5G blockchain space.

B. Without the use of NFTs

The papers of this section are divided into two subcategories, the implementations that use permissioned blockchain and these that use permissionless.

1) Use of permissioned blockchain: The Just-In-Time Resource Allocation (JITRA) [30], developed by Faisal et al., uses blockchain to improve resource allocation in 5G networks. It employs short-term, dynamic service contracts instead of long-term ones. This work implements allocations for the resources through smart contracts in a dynamic fashion. To achieve it, is proposed a Quality of Service (QoS) monitoring system which is responsible of monitoring and enforcing standards through Service Level Agreements (SLAs). Moreover, utilizing Hyperledger Fabric, JITRA interacts with private and permissioned networks corresponding to real-time demand. This proposal describes a system that allocates and reallocates resources in real time to deliver optimal performance and resource utilization. Accordingly, significant challenges are to manage numerous short-term contracts and reduce possible false positives in QoS monitoring. The current approach to resource allocation and network performance remains reliant on static resource allocation techniques. In contrast, JITRA's advantage lies in its ability to adapt resource distribution in real time. While there are advantages, it is a very complex task to handle many dynamic service contracts quickly.

Mensah et al. proposed the Blockchain-Enabled Federated Learning-Based Resource Allocation and Trading [31]. This describes a Federated Learning-based technique for allocating and trading resources, integrated with network slicing and blockchain. A Tendermint-based blockchain was integrated to serve as the mechanism for consensus; as a result, by design, transactions, and data exchanges are secure and privacy-oriented. This enables it to securely reach an agreement quickly, even if some nodes are faulty. Federated learning optimizes resource allocation decisions without compromising data privacy. More specifically, FL allows each network slice tenant to train models locally, so raw data is never shared, maintaining privacy, while it utilizes smart contracts to automate resource actions. This will ensure traceable and verifiable transactions. However, the integration comes with high computational overhead and complexity.

Quang Huy Do et al. proposed the Blockchain Integration in 5G Technologies: SDN and Network Slicing [32]. It focuses on the integration of SDN and blockchain technology to execute network slicing in 5G networks. This work utilizes hyperledger fabric [33] for their operations and transactions for the network resources. SDN brings centralized management and control, whereas smart contracts bring resource allocations. Thus, it provides security against vulnerabilities and privacy issues in network slicing. One of the key challenges is the scalability of blockchain technology in this setting. Current blockchain systems are unlikely to support the kind of 5G networks that will need to process thousands of transactions per second. Additionally, managing a blockchain-integrated SDN environment can be complex and energy-intensive.

Aryal et al. proposed Moving Towards Open Radio Access Networks with Blockchain Technologies [34]. This paper describes an implementation that uses blockchain to enrich O-RAN security and efficiency. The most salient feature is using a Quorum [35] permissioned blockchain platform to develop the framework for guaranteed decentralized control, enabling higher security for O-RAN architectures. Quorum is an enterprise-focused blockchain network which has been built to be a permissioned version of Ethereum to support business needs. This provides immutability and decentralization. On the other hand, scalability and performance issues are challenges that occur with the use of blockchain. Moreover, establishing governance frameworks and preventing network fragmentation are additional hurdles.

2) Use of permissionless blockchain: Afraz et al. from University of Dublin introduced the 5G Network Slice Brokering [36] that describes a blockchain-enabled market mechanism for network slice brokering in 5G networks. It proposes a mechanism of automatically leasing network slices among heterogeneous stakeholders through a decentralized marketplace on the Ethereum blockchain, with the utilization of smart contracts. This approach reduces manual brokering costs, and dynamically creates network slices based on real-time needs and predefined Service Level Agreements (SLA). This implementation guarantees tamper-proof records through a smart contract and decentralized ledger technology, fostering trust and eliminating the need for a central authority. Even though there are advantages, it needs to address issues like vast demand for computing resources, increased network traffic, limitations in scalability, and integration complexities.

Table I reveals the comparison of architectures of the different blockchain-based slicing solutions, highlighting key attributes and their respective advantages and disadvantages. It utilizes distinct blockchain technologies, such as Hyperledger Fabric, Ethereum, Tendermint etc. with some incorporating AI models like Federated Learning (FL) and GPT-3.5. The table details their position on integrating NFT, which is crucial for the slicing and the implementation of a marketplace, and the kind of slicing it supports: on-chain, off-chain, or both. The advantages range from scalability, transparency, efficiency, and robust security feature sets to dynamic trading ability for slices. Their setbacks may include issues in high computation overhead, complexity, privacy risks, restriction of scalability, and latency problems. This comparative analysis facilitates a nuanced understanding of the trade-offs involved in implement-

Authors	Technology	Blockchain	AI Model	NFT support	Slicing	Advantages	Disadvantages
Faisal [30]	JITRA	H. Fabric	-	-	●	Scalability	Possible false positives
Afraz [36]	5G Slice Brokering	Ethereum	-	-	●	Transparency, automation	Privacy concerns, scalability issues
Mensah [31]	Blockchain-Enabled FL	Tendermint	FL	-	●	Efficiency	High computational overhead
Bandara [24]	Kaputa	Moose	-	✓	○	Transparency, dynamic slice trading	Complex management, data overhead
Foytik [26]	SliceGPT	Ethereum	GPT-3.5	✓	●	Efficiency	High computational needs, scalability limits
Huy Do [32]	Blockchain SDN	H. Fabric	-	-	⊜	Security	Complexity, latency issues, high oper. costs
Aryal [34]	Open RAN	Quorum	-	-	●	Confidentiality, access control	Scalability issues, storage/latency issues
Shetty [27]	Blockchain NFT FL UAV	Cosmos	FL	✓	⊜	UAV, provenance tracking	High comp. overhead, possible false positives
Michoulis [29]	FraMark	H. Fabric	-	✓	●	Dynamic resource alloc. with fractional NFTs	High computational needs

TABLE I: Comparative analysis of blockchain-enabled network slicing solutions.

✓= NFT support, ● = Slicing supported on-chain, ○ = Slicing supported off-chain, ⊜ = Slicing support on-chain and off-chain

ing blockchain-based network slicing solutions.

IV. Main findings and remaining challenges

A. Main findings

The integration of blockchain technology with 5G networks presents transformative benefits, improves the efficiency, security, and flexibility of network slicing. This paper has surveyed a collection of possible ways and frameworks that express such benefits.

Firstly, the use of blockchain-enabled network slicing has proven to significantly enhance resource management and allocation. The use of NFTs in the representation of the network slices has been proven to be effective, among other strategies. NFTs encapsulate detailed information regarding the network slice like configuration, history of its use, and origin of the resources. It makes the handling and trade of resources decentralized, transparent, and efficient within a marketplace. Moreover, it enables real-time resource allocation to elevate effective resource use and increase flexibility. Smart contracts, automate the orchestration and management of network slices, thereby reducing manual intervention and the likelihood of errors, while simultaneously improving service reliability and consistency.

In addition to this, the incorporation of new technologies like AI and machine learning, coming together, creates benefits. Combining AI, edge computing, and machine learning with blockchain can make network operations more efficient. Increased usage of data-driven analytics will help in the uptake of AI-powered insights and will further provide help in optimizing network slices. Moreover, federated learning scales the security and privacy of the network. With the use of federated learning, data stays local, and only the updates on the models are carried, hence it presents the advantage of data privacy added to the described confidentiality and security of models.

B. Remaining challenges

1) Integration complexity: Blockchain integration over 5G infrastructures has a technical difficulty. Combining the various technologies in AI, Edge computing, and Federated learning adds extra complexity. The improved interoperability and the dynamically changing requirement of network slices across the multiple domains created by the stakeholder make the hurdle. This includes optimizing blockchain operations specifically for speed of transactions, effective use of smart contracts, and causing minimum disruption to the network services within a 5G context.

2) Cybersecurity: Cybersecurity has also been a concern in integrating blockchain in the 5G networks. Relevant papers [32] [31] discuss comprehensive cybersecurity measures, including encryption, secure access control, and decentralized security protocols. On the other hand, some papers dwell more on the theoretical upside of the blockchain, leaving out most of the cybersecurity details which may spell vulnerability. It is necessary to have security in all implementations to be safe against threats like data breaches, hacking, and unauthorized access.

V. Conclusion

This survey highlights the integration of 5G networks and blockchain technology, emphasizing improvements in efficiency, security, and network management through innovative technological solutions such as network slicing and NFTs.

The incorporation of blockchain systems into the telecommunication industry presents a viable solution with promising results offering new features and enhancing existing solutions. However, the practical application of this integration includes challenges such as high complexity in the network architecture and stringent cybersecurity requirements.

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